

SIDDHANT ACHARYA

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 5+ years of experience architecting and delivering high-performance, cloud-native enterprise applications using Java, Spring Boot, and AWS. Proven track record building event-driven microservices, RESTful APIs, and distributed systems at scale — including production deployments serving 2,000+ internal users with 99.9% uptime. Hands-on expertise integrating AI/ML and Generative AI solutions: RAG pipelines (LangChain, Spring AI), LLM-powered automation, and ML-based predictive analytics that drove multi-million-dollar business impact. AWS Certified Solutions Architect with deep experience in Kubernetes, Docker, CI/CD, and Agile delivery.

TECHNICAL SKILLS

Programming Languages: Java (8/11/17/21), Python, Kotlin, JavaScript, TypeScript, SQL, C#

Frameworks & Libraries: Spring Boot, Spring AI, Spring MVC, Spring Data JPA, Hibernate, Node.js, ReactJS, AngularJS, Pandas, NumPy, Scikit-learn

AI/ML & GenAI: LLM Integration, OpenAI API, Spring AI MCP, LangChain, RAG Pipelines, Prompt Engineering, Hugging Face, Ollama

Cloud & DevOps: AWS (EC2, S3, Lambda, SNS, SQS, Glue, IAM, CloudFormation, CloudWatch), GCP, Azure, Docker, Kubernetes, Terraform, Jenkins, Azure DevOps, Octopus Deploy

Web & API: REST APIs, GraphQL, SOAP, gRPC, WebSockets, Apache Kafka, OpenAPI/Swagger

Databases: PostgreSQL, SQL Server, Oracle, MongoDB, DynamoDB, H2, pgvector

Tools & Concepts: Git, IntelliJ IDEA, Postman, Splunk, Grafana, Maven, CI/CD, OOP, DSA, OAuth2, Microservices, HLD/LLD

WORK EXPERIENCE

Software Engineer — Georgia-Pacific (Koch Industries)

Atlanta, GA | Jan 2023 – May 2025

Java · Spring Boot · AngularJS · AWS · Docker · Kubernetes · SQL · Azure DevOps

- Refactored and optimized full-stack modules across ProSearch (Java, Spring Boot) and GPXpress (Java, Spring Boot, SQL) for 560+ users, improving frontend performance by 35% and reducing CSR order resolution time by 25%.
- Implemented critical backend features for the Commissions platform (Java, AngularJS, Docker, Kubernetes), contributing to an estimated 4% reduction in commission overpayments with multi-million-dollar revenue impact; automated deployments via Azure DevOps pipelines, cutting release cycle time by 40%.
- Built and enhanced key features for Tartan (Java, AngularJS, Docker, Splunk), integrating LLM-powered document summarization via Azure OpenAI Service to automate processing of operational reports for 2,000+ internal users, reducing manual review time by 50% and improving decision turnaround by 30%.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline using LangChain and Azure Cognitive Search to surface relevant policy and product knowledge within internal tools, achieving a 40% reduction in agent lookup time and improving first-contact resolution rates by 18%.
- Built predictive analytics integrations for What-If Analysis (Java, Spring Boot) by incorporating ML-based forecasting models, enabling business stakeholders to simulate pricing scenarios with 92% accuracy and reducing manual analysis effort by 3 hours per analyst per week.
- Designed secure alerting and consent management systems (AWS SNS, IAM, Secrets Manager) with event-driven workflows (AWS SNS/SQS), achieving 99.9% system uptime and ensuring compliance with data governance policies across 5+ integrated enterprise systems.
- Mentored a junior developer on engineering best practices, AWS deployments, and AI integration patterns; drove unit and integration test coverage to 80%+, reducing production incidents by 15% over two consecutive quarters.

AWS Intern — Georgia-Pacific (Koch Industries)

Atlanta, GA | May 2022 – Aug 2022

AWS (EC2, S3, CloudFormation, CloudWatch, SNS, IAM)

- Engineered a cloud-based infrastructure monitoring dashboard using AWS CloudFormation, EC2, and S3 to track real-time SAP instance metrics, reducing manual status checks by 60% and improving infrastructure visibility for a team of 15+ engineers.
- Automated billing alerts and security notifications with Amazon SNS and CloudWatch alarms, decreasing average incident response time by 30% and proactively flagging 12+ potential cost overrun events during the internship period.
- Authored reusable CloudFormation templates for provisioning standardized AWS environments, cutting new environment setup time from 4 hours to under 30 minutes and enabling consistent infrastructure-as-code practices across the team.

Software Engineer — Dots Info Systems Pvt. Ltd.

Mumbai, India | Jan 2020 – Mar 2021

ReactJS · TypeScript · HTML/CSS · REST APIs · OpenAPI/Swagger

- Spearheaded development of mobile-first web applications using ReactJS, TypeScript, and HTML/CSS for 3 enterprise clients, boosting mobile user engagement by 25% and achieving responsive design compliance across 10+ device breakpoints.
- Crafted dynamic, reusable UI component libraries with CSS animations and ReactJS best practices, reducing front-end development time for new features by 30% and contributing to a measurable decrease in user bounce rates across client portals.
- Integrated frontend components with RESTful APIs and implemented lazy loading and caching strategies, improving average page load time by 1.8 seconds and driving a 20% increase in repeat user visits month-over-month.
- Collaborated with backend engineers to define and document RESTful API contracts using OpenAPI (Swagger), reducing integration defects by 22% and cutting cross-team debugging time by an estimated 5 hours per sprint.

Software Engineer — Star JaiSi

Mumbai, India | Jan 2019 – Jan 2020

JavaScript · ReactJS · TypeScript · HTML · CSS

- Revamped 5 client websites by building custom, mobile-responsive interfaces with ReactJS, TypeScript, HTML, and CSS, contributing to a 30% increase in client sales through improved customer engagement and conversion-optimized layouts.
- Applied modern SEO techniques and semantic HTML practices to optimize website visibility for 3 key business domains, driving a 25% uplift in organic traffic within 6 months of deployment.
- Integrated interactive UI features and implemented cross-browser compatibility fixes across Chrome, Firefox, and Safari, reducing user-reported UI issues by 35% and improving overall customer satisfaction scores for commercial websites.

RECENT PROJECTS

DocQA — RAG Document Q&A API · [Link](#)

Java 21 · Spring Boot 3.3 · Spring AI 1.0 · Ollama · PostgreSQL/pgvector · Docker

- Reduced LLM infrastructure cost to \$0 by architecting a fully local RAG pipeline using Ollama (Llama3.2 + nomic-embed-text) with Spring AI 1.0, replacing cloud API dependency while maintaining 100% answer accuracy across all test queries on real resume PDF data.
- Achieved sub-450ms semantic retrieval latency by implementing PostgreSQL pgvector with HNSW indexing and 768-dim embeddings via token-aware PDF chunking (PagePdfDocumentReader + TokenTextSplitter) in a Docker-composed vector store.
- Engineered provider-portable Spring AI ChatClient abstraction enabling zero-refactor swap between Ollama, OpenAI, and Anthropic via single pom.xml dependency change; hardened with Bean Validation, global exception handling, and Actuator health/metrics endpoints.

AI FinOps MCP Server · [Link](#)

Java 21 · Spring Boot 3.3 · Spring AI 1.0 MCP · Spring Data JPA · H2

- Identified 94% AI cost reduction opportunity (\$3.15 → \$0.07/month per team) by engineering a model migration ROI calculator, analyzing 500 usage records across 5 teams and 4 LLM providers using custom JPQL aggregation queries.
- Designed a Spring Boot MCP (Model Context Protocol) server exposing 6 strongly-typed @Tool methods, enabling LLM clients to query AI spending data conversationally — cutting analyst query time from manual REST calls to sub-second natural language responses.
- Architected zero-dependency self-contained service (H2 in-memory, no Docker, no API keys) with 5 custom JPQL aggregations for group-by cost rollups, outlier detection, and period-over-period analysis; auto-seeds 500 records on startup for immediate demo capability.

EDUCATION

M.S. in Engineering Management — Trine University, Reston, VA

Oct 2025 – Dec 2027

M.S. in Computer Science — Missouri State University, Springfield, MO

Jan 2021 – Dec 2022

B.E. in Information Technology — University of Mumbai, Mumbai, India

Aug 2014 – Jun 2018

CERTIFICATIONS

- AWS Certified Cloud Practitioner | Amazon Web Services
- AWS Certified Solutions Architect – Associate | Amazon Web Services